

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Evaluated and validated **quantitative outputs** from enterprise systems by analyzing **production data** across **PostgreSQL, MySQL, Oracle** databases; designed **ad-hoc SQL queries** and statistical checks to assess data integrity and model assumptions in real-time monitoring dashboards using **Splunk** and **Dynatrace**.
- Designed and implemented **Python-based data pipelines** and **ETL workflows** for automated reporting and statistical aggregation; wrote **end-to-end analytical code** using **pandas** and **NumPy** to process, validate, and transform data for downstream analysis and decision-making.
- Developed **REST APIs** and microservices in **Java Spring Boot** supporting quantitative feature work and bug fixes; demonstrated strong coding practices through **Git** version control, peer code reviews, and **CI/CD pipelines** via **Jenkins**.
- Conducted **statistical analysis** and **experiment validation** as part of production support—performing hypothesis testing, root-cause analysis of P1/P2 incidents, and SLA-driven problem-solving to ensure system reliability and data accuracy.
- Built and maintained **Docker** containerized applications on **AWS** infrastructure (**EC2, S3, Lambda, CloudWatch**); automated operational tasks using **Linux shell scripting** and **Python** to improve team efficiency and reduce manual work by standardizing deployment processes.
- Completed **MSc in Data Analytics** alongside full-time engineering role, demonstrating practical expertise in statistical methods, predictive modeling, and research methodology; earned **IELTS Band 7/9** and completed supplementary certifications in **Python Programming** (University of Michigan) and **Full Stack Java Spring Boot** development.

## SKILLS

**Quantitative & Statistical Analysis** – hypothesis testing, statistical inference, data validation, A/B testing, root-cause analysis, SLA management.

**Data Engineering & Pipelines** – ETL workflows, data transformation, automated reporting, pandas, NumPy, SQL query optimization.

**Programming Languages** – Python, Java, SQL, Shell scripting.

**Frameworks & Tools** – Spring Boot, REST APIs, Flask, FastAPI, Docker, Jenkins, Git.

**Cloud & Infrastructure** – AWS (EC2, S3, Lambda, CloudWatch), Linux, monitoring (Splunk, Dynatrace).

**Databases** – PostgreSQL, MySQL, Oracle, schema design, data integrity validation.

**Soft Skills & Credentials** – technical writing, code documentation, Agile/Scrum, IELTS Band 7, MSc Data Analytics, BE Computer Science.

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of  
Ireland, Dublin**

**02/2026**

**BACHELOR OF ENGINEERING in Computer Science – SNS College  
of Technology, Coimbatore, India**

**01/2021**

## PROJECTS

### **Production Data Validation & Quantitative Analysis Framework (HCL Technologies)**

- Designed **Python scripts** using **pandas** and **NumPy** to validate **statistical properties** of production data flowing through enterprise applications; automated detection of data anomalies and schema inconsistencies across **PostgreSQL** and **MySQL** environments.
- Built **SQL-based analytical queries** for ad-hoc hypothesis testing and root-cause analysis during P1/P2 incident response; evaluated correctness of business logic through statistical inference and SLA-driven performance benchmarking.
- Integrated validation workflows with **Splunk** and **Dynatrace** dashboards to provide real-time feedback on system reliability; documented findings in clear technical reports for stakeholder review and model improvement.

### **Automated ETL & Reporting Pipeline (HCL Technologies)**

- Engineered **Python-based ETL pipeline** to extract, transform, and load quantitative data from multiple sources; scheduled automation via **cron jobs** and **Lambda functions** on **AWS** to reduce manual reporting effort by **60%**.
- Wrote **end-to-end analytical code** with **Flask** microservice exposing insights via **REST API**; deployed as **Docker container** on **EC2** with continuous integration via **Jenkins** pipelines and **Git** version control.
- Collaborated with data science and business teams to iterate on pipeline logic; provided clear technical documentation and code reviews to ensure maintainability and statistical correctness of output datasets.

## ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)